

Give the Duck a Qwacker

Project Overview

Project Name: Give the Duck a Qwacker

Type: Quiz Game (Single-player)

Development Methodology: Waterfall

Tools & Technologies: Python, Tkinter, Pillow

Objective:

Develop a fully functional quiz game where players answer trivia questions to feed a duck crackers, track their scores, time the game, and compete via a leaderboard.

Scope:

- Start screen with menu options inclusive of instructions (in a pop out), choice to view a leaderboard (in a pop out), and a start button
- Gameplay with questions, multi-choice answer buttons, scoring, round timing, and a duck that eats crackers with every correct answer
- Game Over screen with leaderboard (in a pop out), play again and exit buttons
- Local storage of scores

The client has provided a brief for a quiz game centred around feeding a duck by answering questions correctly. These are the functional and non-functional requirements of the new game app.

Functional Requirements

Requirement	Description
Player Name	Players can enter a name or nickname (real or pseudonym) to track leaderboard position.
Start Page	Players are welcomed with a start screen with options to start the game, view the leaderboard or view instructions.
Instructions	The game play instructions are available on the start page.
Questions	The questions must give multiple choice answer with 3 options.
Answer Input and Validation	Players can click on answers.
Correct/Wrong Answers	A message must shown for correct or incorrect answers.
Score Tracking	The player's score is tracked and displayed during gameplay.
Cracker Feeding Animation	The duck is to be animated to move automatically and eat a cracker.
Question Progression	The next question is to be loaded automatically after duck moves.
Game Over Screen	To display final score and time and give options to play again, exit or view leaderboard.
Leaderboard	A leaderboard displaying quickest time can be viewed in a pop out showing player names and scores.
Game Timer	The game will be timed to answer 10 questions in under 'x' time to get the duck home to his pond. If this is not achieved then game over screen to show.
Responsive Controls	The game accepts player input via mouse or trackpad.

Non-Functional Requirements

Requirement	Description
Security	Player names/nicknames/pseudonyms are stored without personal details.
Performance	Should run smoothly with animations and quick loading of questions and screens.
Usability	User interface is intuitive and accessible, with clear instructions and visible feedback messages.
Reliability	Should save and accurately display player scores, times and leaderboard without errors.
Scalability	Designed to allow easy addition of new questions, different animal choice, and additional features/min-games in future.
Maintainability	Codebase should be clean and modular to allow future updates, bug fixes, or feature additions easily.
Availability	The game should run consistently on supported devices without crashes or freezes.
Compatibility	The game must work across multiple platforms including Windows, macOS, Android, and iOS with consistent UI.
Responsiveness	UI and controls should respond to inputs with minimal delay.
Accessibility	The game should be playable by users with disabilities, including support for screen readers and have highly visible text contrast.